

Automated Processing of Climate Projection NetCDF Files in R

Technical Documentation

This document describes the structure and purpose of an R script designed to automate the processing of climate projection NetCDF (.nc) files containing precipitation data from the Copernicus Climate Data Store (CDS). The workflow extracts daily precipitation values, processes geospatial data, and generates time-series outputs for a defined region (Zuid-Holland, Netherlands). The script implements an ETL (Extract, Transform, Load) pipeline and includes extensive error handling and memory management to allow batch processing of large climate datasets.

Dataset Description

Dataset Source: Copernicus Climate Data Store (CDS) Dataset: Hydrology and Meteorology Derived Climate Projections Key Characteristics: • Variable: Precipitation • Processing type: Bias corrected • Temporal resolution: Daily • Spatial resolution: 5 km • Climate experiment: RCP 8.5 / RCP 2.6 scenario variants • Regional Climate Model: RACMO22E (KNMI, Netherlands) • Global Climate Model: HadGEM2-ES (UK Met Office) • Projection grid: Lambert Azimuthal Equal Area (LAEA Europe – EPSG:3035)

Processing Pipeline Overview

The script follows an ETL data pipeline structure: 1. Extract - Load NetCDF precipitation files year-by-year - Retrieve spatial coordinates and temporal metadata - Load the Zuid-Holland municipal shapefile 2. Transform - Convert NetCDF raster data into SpatRaster objects - Reproject raster layers to EPSG:3035 - Crop and mask raster data using the shapefile boundary - Convert precipitation units from $\text{kg m}^{-2} \text{s}^{-1}$ to mm/day - Compute spatial mean precipitation values 3. Load - Export yearly precipitation time series to CSV - Combine yearly CSV files into a single aggregated dataset - Perform integrity checks to detect missing years

1. Workspace Initialization

The script begins by clearing the R environment using `rm(list = ls())` to avoid object conflicts. Temporary file locations are checked using `tempdir()` to diagnose disk-space related crashes, which are common when working with large NetCDF raster datasets.

2. Library Dependencies

The following R packages are required: • `terra` – raster and spatial data processing • `sf` – shapefile and vector spatial data • `ncdf4` – NetCDF file interaction • `dplyr` – data aggregation and manipulation • `ggplot2` – visualization of processed time series

3. Shapefile Loading and Projection

The municipal shapefile for Zuid-Holland is loaded using `sf::st_read()`. The original coordinate reference system (CRS) is EPSG:28992 (Amersfoort / RD New). The shapefile is reprojected to EPSG:3035 (ETRS89 / LAEA Europe) to match the climate dataset projection and ensure compatibility with raster masking operations.

4. Yearly Batch Processing Loop

A loop iterates over a defined sequence of years (2025–2080). For each year:

- The corresponding NetCDF file is located
- Raster data is loaded via `terra::rast()`
- Metadata is extracted using `ncdf4::nc_open()`
- Time units are converted from seconds since origin to POSIX datetime

Error handling is implemented with `tryCatch()` to prevent a failure in one year from terminating the entire processing loop.

5. Spatial Processing Workflow

Each NetCDF file undergoes the following geospatial operations:

1. Assign CRS EPSG:4326 to raw raster data
2. Reproject raster to EPSG:3035
3. Crop raster to shapefile extent
4. Mask raster using municipal boundaries

These steps ensure precipitation data is spatially restricted to the study area.

6. Unit Conversion

Precipitation data in NetCDF files is stored as $\text{kg m}^{-2} \text{s}^{-1}$. Since 1 kg/m^2 corresponds to 1 mm of rainfall and there are 86,400 seconds in a day, the conversion to mm/day is: `precipitation_mm_day = precipitation * 86400`

7. Time Series Extraction

The spatial mean precipitation value for each raster time step is calculated using: `terra::global(raster, fun = "mean", na.rm = TRUE)`. This produces a daily precipitation time series for the study region. The resulting data frame contains:

- Date
- Precipitation_mm_day

8. CSV Export

Each processed year is exported as a CSV file: `YYYY_precipitation_ts.csv`. This modular storage structure prevents memory overflow and allows intermediate results to persist in case of script interruption.

9. Aggregation of Yearly CSV Files

After processing all years, the script aggregates individual CSV files into a single dataset using `dplyr::bind_rows()`. The resulting file: `Combined_RCP2.6_2025-2080.csv` contains the complete precipitation time series across all years.

10. Data Integrity Check

The script verifies whether all expected years are present by comparing:

- Expected year sequence
- Years detected in the combined dataset

Missing years are printed in the console for troubleshooting.

11. Visualization

A final visualization step uses `ggplot2` to produce a precipitation time series plot. The Date column is converted to `POSIXct` format, and precipitation values are plotted as a line chart to inspect temporal patterns and extreme events.

12. Future Automation Improvements

Planned extensions to the script include:

- Integration of the CDS API for automated dataset downloads
- Direct pipeline integration with Extreme Value Theory models (GPD)
- Python-based replication using machine learning approaches
- More advanced precipitation threshold analysis for extreme rainfall events